



PANDUAN KONFIGURASI ROUTER & SWITCH CISCO

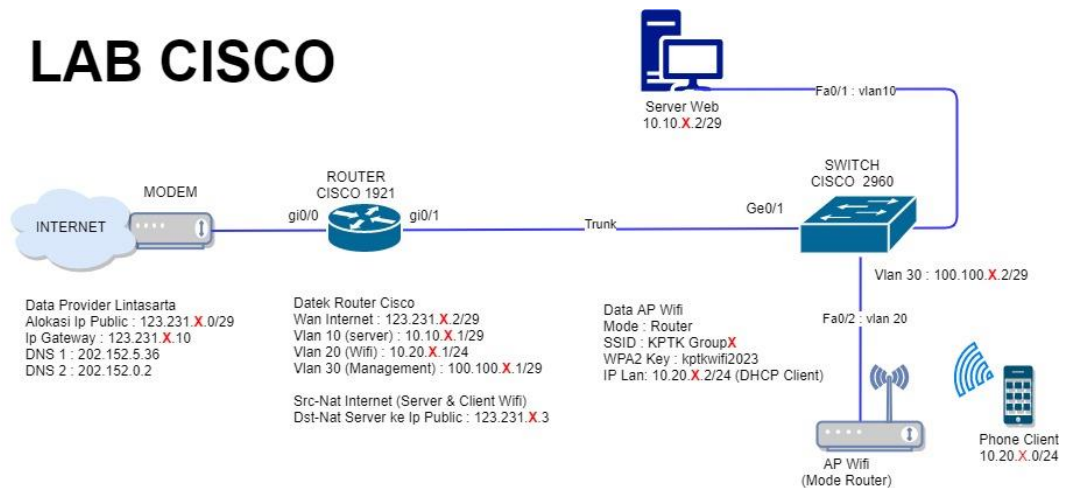


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1. Topologi



Pada topologi diatas, terdapat beberapa Informasi :

- Nilai "X" disesuaikan dengan nomor kelompok group lab.
- Cisco 1921 berfungsi sebagai router atau gateway ke internet.
- Cisco 2960-C berfungsi sebagai switch untuk melewati beberapa vlan.
- AP Wifi untuk distribusi Lan ke Client.
- Terdapat Server Web untuk test Aplikasi.

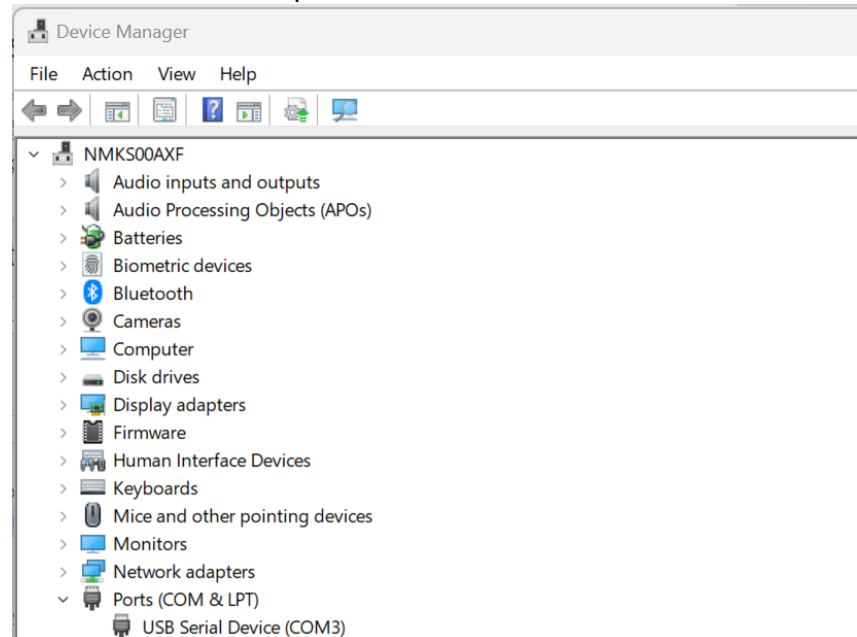
2. Akses Perangkat

Untuk mengakses perangkat router atau switch cisco bisa menggunakan kabel USB Console.

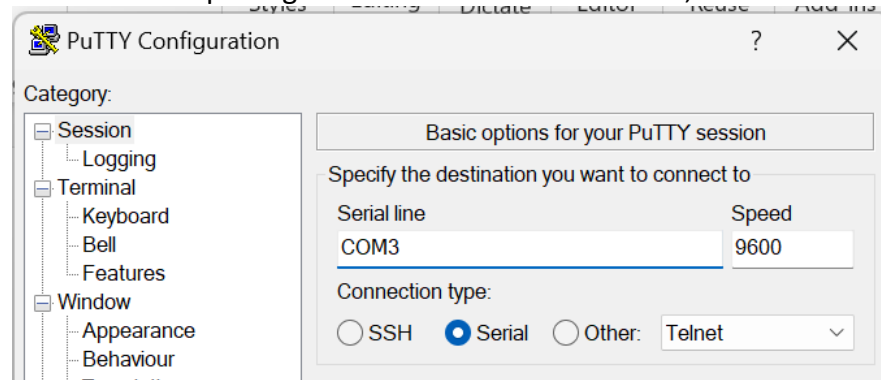
- 1) Hubungkan USB console ke Laptop



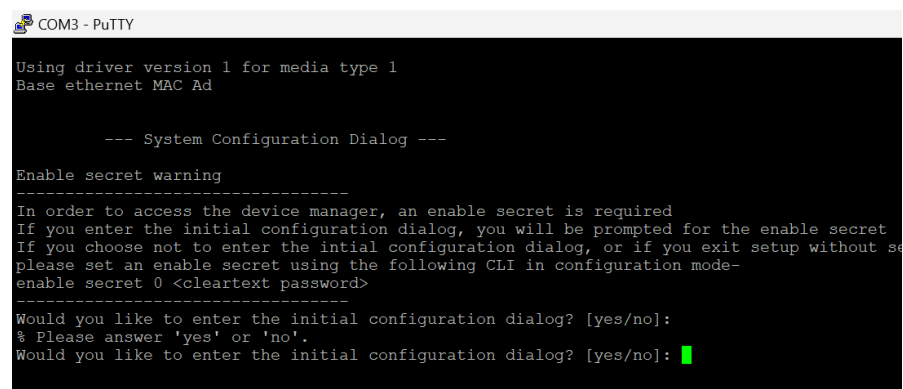
- 2) Cek Pada Device Manager Port USB console cisco. Pada contoh dibawah terlihat USB port berada di COM3.



- 3) Silahkan akses perangkat melalui terminal emulator, misal via PuTTY.



Jika berhasil akan muncul tampilan *Command Line Interface (CLI)* cisco sebagai berikut.



3. Konfigurasi Router Cisco (Contoh Konfigurasi Group1, Group Lain silahkan menyesuaikan sesuai topologi)

- 1) Untuk melakukan konfigurasi router, pastikan anda dalam posisi mode konfigurasi.

router# configure terminal

- 2) Setup nama atau hostname router.

router(config)#hostname ROUTER-GROUP1

- 3) Setup username, password dan secret router

***username admin privilege 15 password admin123
service password-encryption
enable secret admin123***

Catatan :

- Ada beberapa level privilege (tingkat hak akses) yang digunakan untuk mengatur akses dan izin pengguna di cisco. Privilege levels berkisar dari 0 hingga 15, dengan 0 adalah level paling rendah (terbatas) dan 15 adalah level paling tinggi (superuser atau administrator penuh).
- service password-encryption berfungsi untuk mengenkripsi (mengacak) kata sandi yang disimpan dalam konfigurasi perangkat.
- enable secret berfungsi sebagai sandi pengaman untuk melindungi akses ke mode administrator penuh/mode privileged exec.

- 4) Setup line vty agar router bisa ditelnet

***line vty 0 4
login local
transport input telnet
transport output telnet***

Catatan : Line vty adalah virtual interface untuk mengakses/melakukan remote perangkat cisco melalui jaringan. Line vty 0 4, artinya dalam satu waktu bersamaan, maksimal bisa ada 5 koneksi yg terkoneksi ke perangkat (0 - 4 berjumlah 5 dimulai angka 0, 1, 2, 3, dan 4). Maksimalnya ialah 16 koneksi pada saat bersamaan (line vty 0 15)

- 5) Setup snmp server agar bisa dimonitoring oleh NMS (misal community menggunakan "public" dengan hak akses "Read Only")

snmp-server community public RO

- 6) Setup Interface gi0/0 yang mengarah ke Internet dan aktifkan port-nya.

interface GigabitEthernet0/0
no shutdown
description WAN-INTERNET
ip address 123.231.1.2 255.255.255.248
ip nat outside

Catatan : ip nat outside digunakan untuk menandai interface yang terhubung ke sumber daya jaringan eksternal sebagai "outside" dalam proses *Network Address Translation (NAT)*.

- 7) Berikan Deskripsi Interface Trunk gi0/1 dan aktifkan port-nya.

interface GigabitEthernet0/1
no shutdown
description TRUNK-SWITCH
no ip address

- 8) Buat sub interface (vlan) untuk masing-masing koneksi ke arah LAN

Sub Interface untuk Lan-Server (vlan10)

interface GigabitEthernet0/1.10
description LAN-SERVER
encapsulation dot1Q 10
ip address 10.10.1.1 255.255.255.248
ip nat inside

Sub Interface untuk Lan-Wifi (vlan20)

interface GigabitEthernet0/1.20
description LAN-WIFI
encapsulation dot1Q 20
ip address 10.20.1.1 255.255.255.0
ip nat inside

Sub Interface untuk Lan-Manajemen (vlan30)

interface GigabitEthernet0/1.30
description LAN-MANAJEMEN
encapsulation dot1Q 30

ip address 100.100.1.1 255.255.255.248

- 9) Buat routing default mengarah ke ip wan gateway untuk semua ip.
ip route 0.0.0.0 0.0.0.0 123.231.1.1 name ROUTING-INTERNET
- 10) Buat Access-list ip lan yang akan di nat ke Internet

access-list 1 permit 10.20.1.0 0.0.0.255
access-list 1 permit 10.10.1.0 0.0.0.7

- 11) Buat IP Nat Pool ip wan agar lan bisa terkoneksi ke Internet dan rule nat-nya.

ip nat pool INTERNET 123.231.1.2 123.231.1.2 netmask
255.255.255.248
ip nat inside source list 1 pool INTERNET overload

ip nat outside source static 123.231.1.3 10.10.1.2

Jika ingin spesifik port tertentu saja. Bisa menggunakan opsional command berikut (jika nat full sudah dijalankan, command berikut tidak perlu)

ip nat outside source static tcp 123.231.1.3 443 10.10.1.2 443
extendable

- 12) Setelah konfigurasi router, jangan lupa di simpan konfigurasinya dengan perintah :

write
Atau
write memory

- 13) Lakukan test koneksi ke gateway dan internet (test ping)

```
ROUTER-GROUP1#ping 123.231.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 123.231.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
ROUTER-GROUP1#ping 8.8.8.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/41/44 ms
ROUTER-GROUP1#
```

14) Test Koneksi ke Internet dengan source lan

```
ROUTER-GROUP1#ping 8.8.8.8 so 10.20.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
Packet sent with a source address of 10.20.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/40/44 ms
ROUTER-GROUP1#ping 8.8.8.8 so 10.10.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
Packet sent with a source address of 10.10.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/41/44 ms
```

15) Cek status Nat

```
ROUTER-GROUP1#sh ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 123.231.1.2:8      10.10.1.1:8       8.8.8.8:8          8.8.8.8:8
icmp 123.231.1.2:7      10.20.1.1:7       8.8.8.8:7          8.8.8.8:7
```

16) Cek Ip yang terkoneksi ke Cisco

```
ROUTER-GROUP1#sh ip arp
Protocol Address          Age (min)  Hardware Addr  Type   Interface
Internet 10.10.1.1          -          e4aa.5d9b.c141 ARPA   GigabitEthernet0/1.10
Internet 10.20.1.1          -          e4aa.5d9b.c141 ARPA   GigabitEthernet0/1.20
Internet 100.100.1.1        -          e4aa.5d9b.c141 ARPA   GigabitEthernet0/1.30
Internet 123.231.1.1        1          6c3b.6b4a.fa69 ARPA   GigabitEthernet0/0
Internet 123.231.1.2        -          e4aa.5d9b.c140 ARPA   GigabitEthernet0/0
```

4. Konfigurasi Switch Cisco

17) Setup nama atau hostname router.

```
router#conf t
router(config)#hostname SWITCH-GROUP1
```

18) Setup username, password dan secret router

```
username admin privilege 15 password admin123
service password-encryption
enable secret admin123
```

19) Setup line vty agar router bisa ditelnet

```
line vty 0 4
login local
transport input telnet
transport output telnet
```

- 20) Setup snmp server agar bisa dimonitoring oleh NMS (misal community menggunakan “public” dengan hak akses “Read Only”

snmp-server community public RO

- 21) Setup port trunk yang mengarah ke Router Cisco

***interface GigabitEthernet0/1
no shutdown
description TRUNK-ROUTER
switchport trunk allowed vlan 10,20,30
switchport mode trunk***

- 22) Setup Port akses yang mengarah ke LAN

***interface FastEthernet0/1
no shutdown
description PORT-SERVER
switchport access vlan 10
switchport mode access***

***interface FastEthernet0/2
no shutdown
description PORT-WIFI
switchport access vlan 20
switchport mode access***

- 23) Setup vlan manajemen agar switch bisa diremote

***interface Vlan30
no shutdown
description MANAGEMENT
ip address 100.100.1.2 255.255.255.248***

- 24) Tambahkan Routing Default

ip default-gateway 100.100.1.1

- 25) Setelah konfigurasi switch, jangan lupa di simpan konfigurasinya dengan perintah :

write atau
write memory

26) Melihat status ip address dan interface

```
SWITCH-GROUP1#sh ip int bri
```

Interface	IP-Address	OK?	Method	Status	Protocol
Vlan1	unassigned	YES	NVRAM	up	down
Vlan30	100.100.1.2	YES	NVRAM	up	up
FastEthernet0/1	unassigned	YES	unset	up	up
FastEthernet0/2	unassigned	YES	unset	up	up
FastEthernet0/3	unassigned	YES	unset	down	down
FastEthernet0/4	unassigned	YES	unset	down	down
FastEthernet0/5	unassigned	YES	unset	down	down
FastEthernet0/6	unassigned	YES	unset	down	down
FastEthernet0/7	unassigned	YES	unset	down	down
FastEthernet0/8	unassigned	YES	unset	down	down
GigabitEthernet0/1	unassigned	YES	unset	up	up

27) Melihat deskripsi interface

```
SWITCH-GROUP1#sh int desc
```

Interface	Status	Protocol	Description
Vl1	up	down	
Vl30	up	up	MANAGEMENT
Fa0/1	up	up	PORT-SERVER
Fa0/2	up	up	PORT-WIFI
Fa0/3	down	down	
Fa0/4	down	down	
Fa0/5	down	down	
Fa0/6	down	down	
Fa0/7	down	down	
Fa0/8	down	down	
Gi0/1	up	up	TRUNK-ROUTER
Gi0/2	down	down	

28) Melihat interface trunk

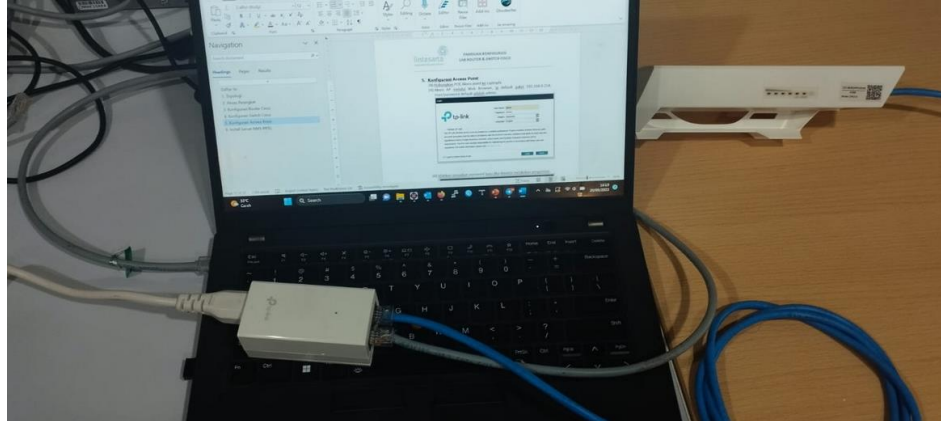
```
SWITCH-GROUP1#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/1	on	802.1q	trunking	1
Port	Vlans allowed on trunk			
Gi0/1	10,20,30			
Port	Vlans allowed and active in management domain			
Gi0/1	10,20,30			
Port	Vlans in spanning tree forwarding state and not pruned			
Gi0/1	10,20,30			

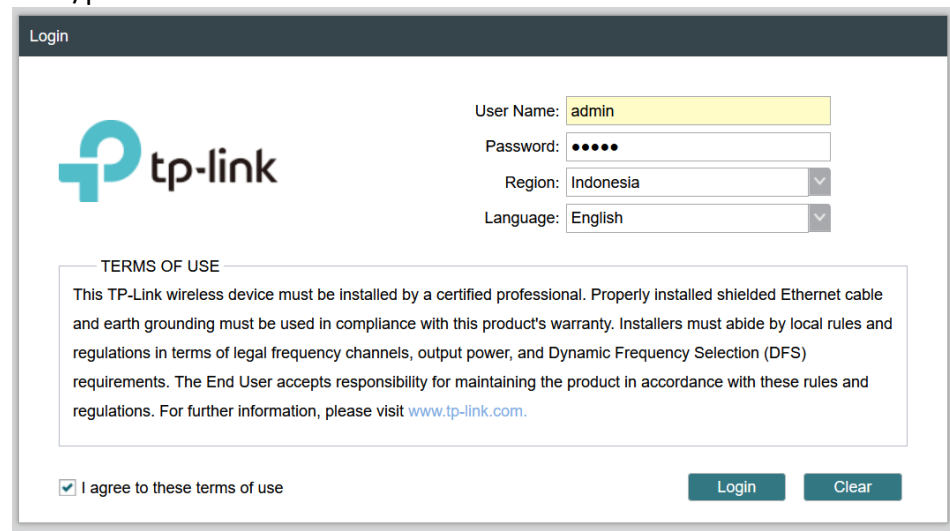
5. Konfigurasi Access Point

29) Hubungkan Adaptor (POE) Akses point/Wifi sebagai berikut

- Port LAN ke Laptop
- Port POE ke Access Point
- Setting IP laptop 192.168.0.1/24 atau bisa diset otomatis/DHCP



30) Akses AP melalui Web Browser, ip default yakni 192.168.0.254. User/password default adalah admin.



31) Silahkan sesuaikan password baru jika diminta melakukan pergantian.

Change Password



New User Name:
New Password:
Confirm Password:

TERMS OF USE

It is recommended to change the device user name and password from its default settings.

Finish

Clear

32) Konfigurasi terlebih dahulu pada menu Wireless.

QUICK SETUP

STATUS

NETWORK

WIRELESS

MANAGEMENT

SYSTEM

Basic Wireless Settings

Region:

Mode:

Channel Width:

Max TX Rate:

Channel/Frequency:

☐ Channel Shifting

Transmit Power: dBm

MAXtream: ☐ Enable

Spectrum Analysis

Wireless AP Settings

Wireless Radio: ☒ Enable

SSID:

☒ Enable SSID Broadcast

Security Mode:

Version:

Encryption:

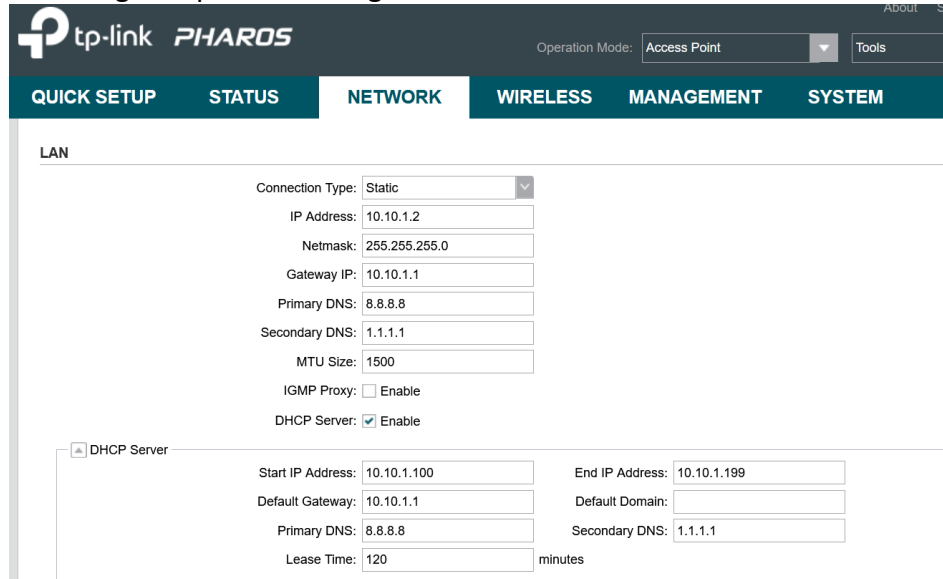
PSK Password:

☒ Show

Group Key Update Period:

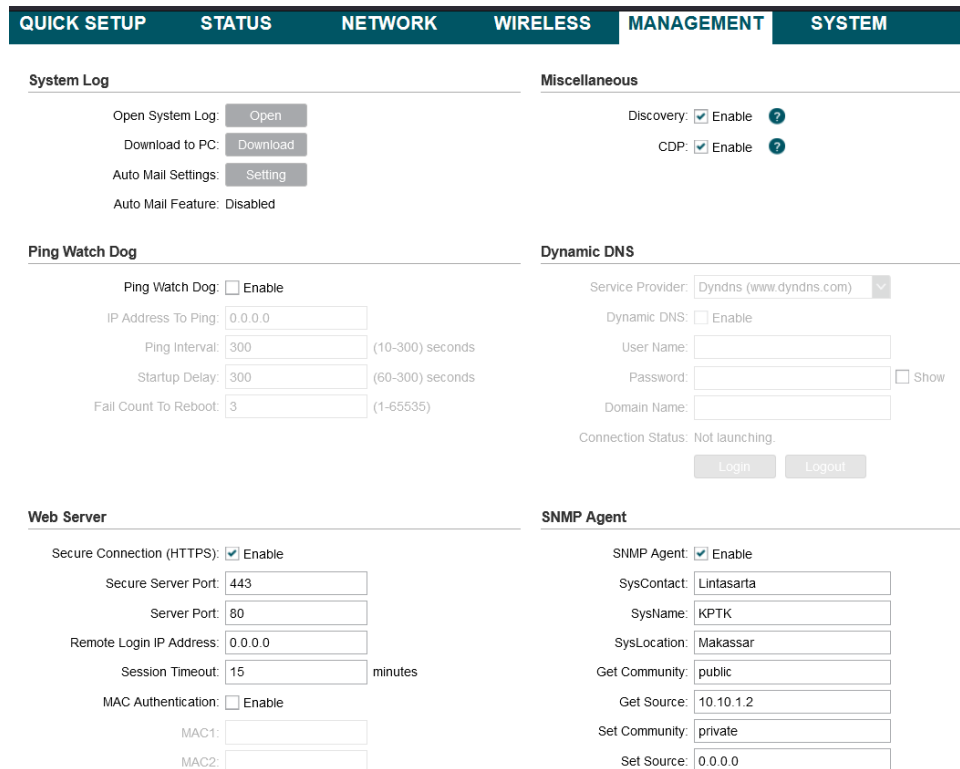
seconds. (0 means no update.)

33) Selanjutnya konfigurasi Ip Address AP sesuai network ip gateway router agar dapat terhubung satu sama lain.



Catatan : DHCP Server disetup pada Access Point (AP). Jika telah ada DHCP Server di Cisco, maka DHCP Server AP di disable.

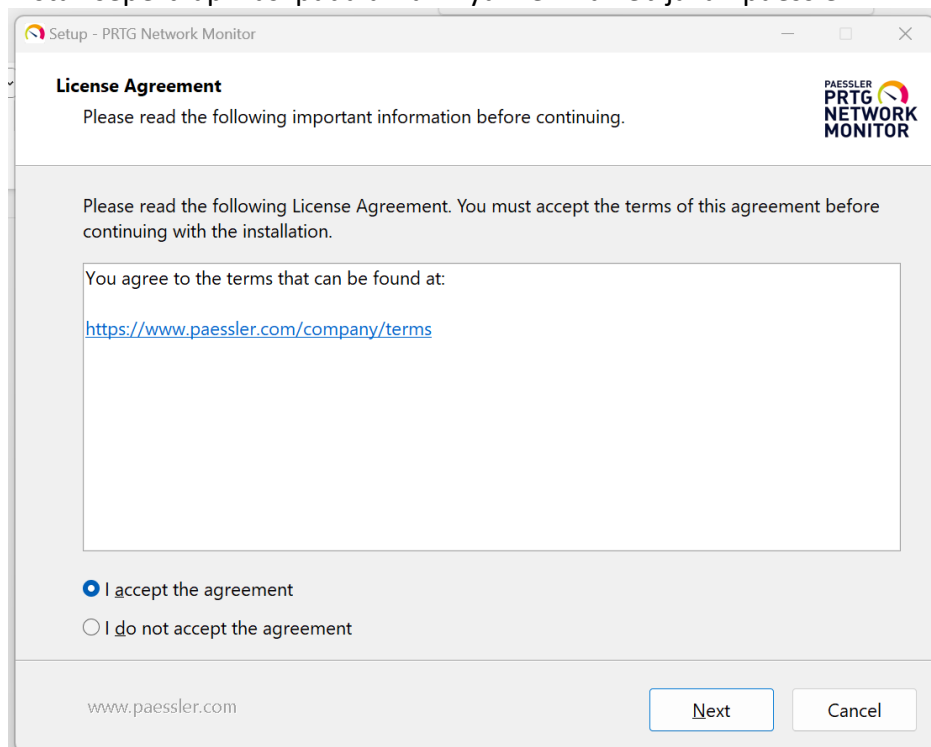
34) Jangan lupa aktifkan SNMP agent di Menu Management. Agar bisa dimonitor.



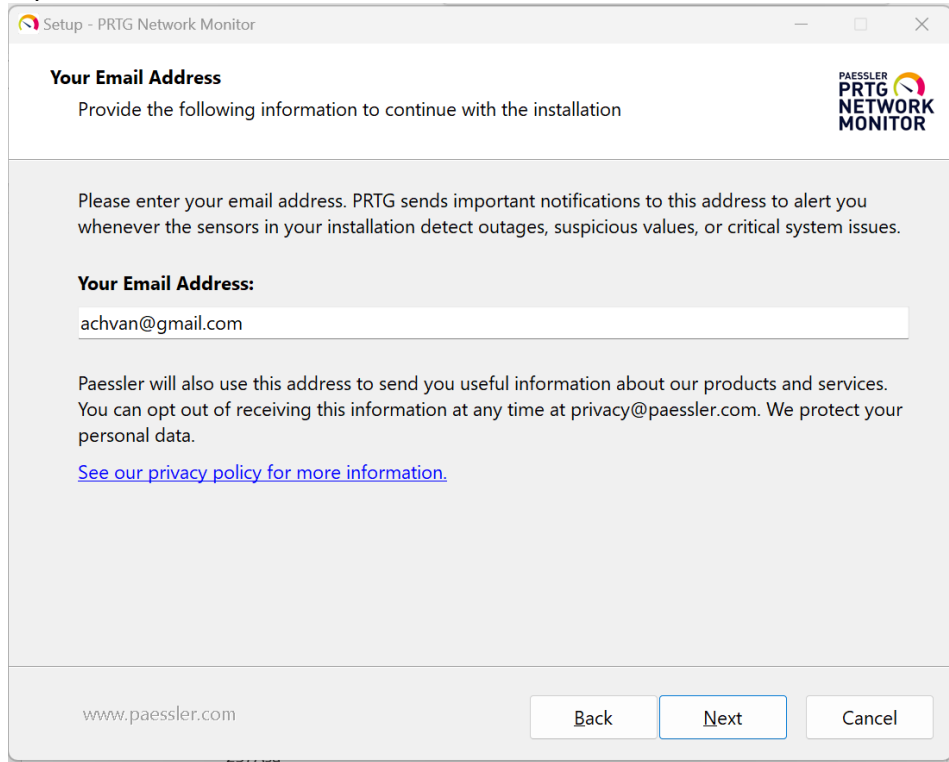
35) Selanjutnya bisa dilakukan pengetesan koneksi dari Wifi Client.

6. Install Server NMS PRTG (Opsional)

- 36) Pada Lab berikut kita akan menggunakan PRTG (Paessler PRTG) sebagai server Network Monitoring System (NMS). Situs resmi bisa diakses melalui url : <https://www.paessler.com>
- 37) PC/Laptop yang dilakukan instalasi PRTG :
- Dipastikan sudah terkoneksi LAN ke Mikrotik port ether2
 - Setting ip sesuai alokasi, dalam hal ini 10.10.X.2/29
- 38) Untuk download aplikasi prtg, bisa di download dari situs paessler.
<https://www.paessler.com/prtg/prtg-network-monitor>
- 39) Install seperti aplikasi pada umumnya. Terima kebijakan paessler.

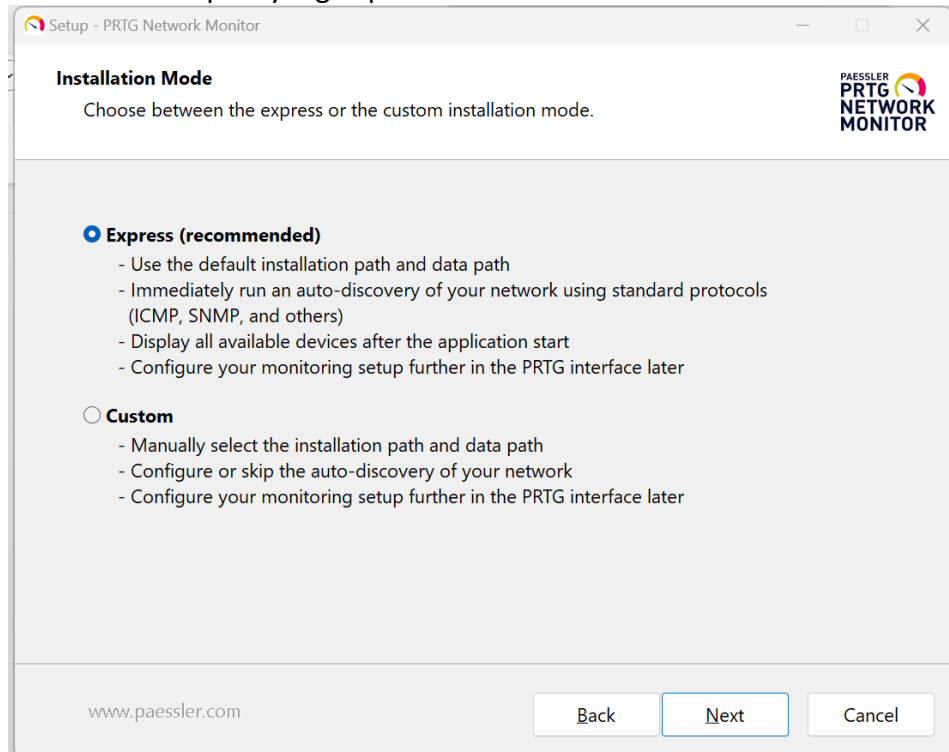


40) Input Alamat email



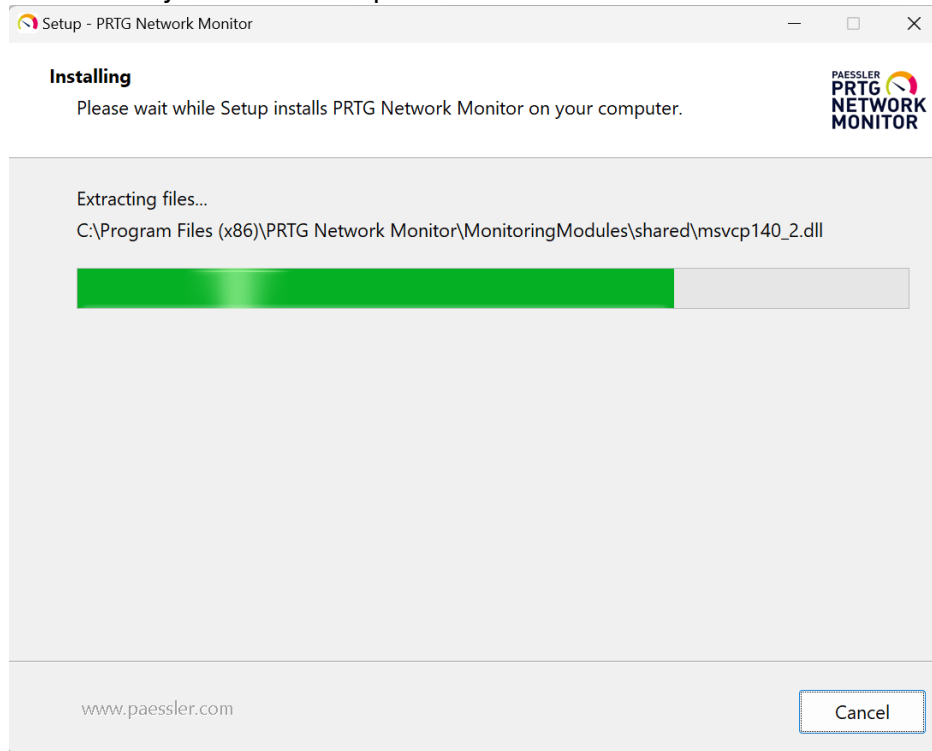
The screenshot shows the 'Setup - PRTG Network Monitor' window. The title bar includes the PRTG logo and the text 'Setup - PRTG Network Monitor'. The main content area is titled 'Your Email Address' and contains the following text: 'Provide the following information to continue with the installation'. Below this, a paragraph states: 'Please enter your email address. PRTG sends important notifications to this address to alert you whenever the sensors in your installation detect outages, suspicious values, or critical system issues.' A text input field is labeled 'Your Email Address:' and contains the email 'achvan@gmail.com'. Below the input field, another paragraph states: 'Paessler will also use this address to send you useful information about our products and services. You can opt out of receiving this information at any time at privacy@paessler.com. We protect your personal data.' A link is provided: '[See our privacy policy for more information.](#)'. At the bottom, there is a footer with 'www.paessler.com' and three buttons: 'Back', 'Next' (highlighted with a blue border), and 'Cancel'.

41) Mode Instalasi pilih yang express

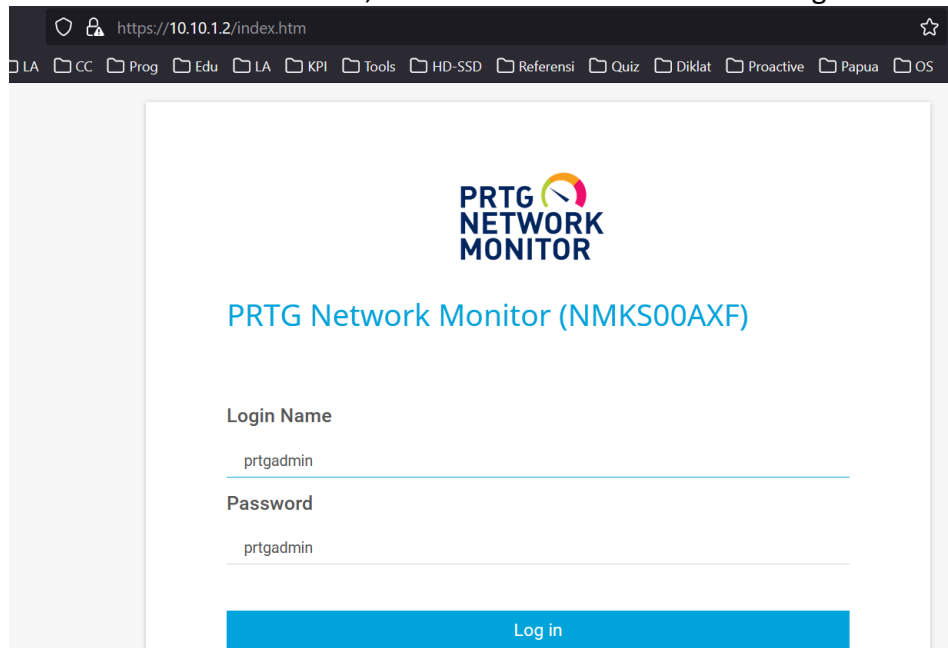


The screenshot shows the 'Setup - PRTG Network Monitor' window. The title bar includes the PRTG logo and the text 'Setup - PRTG Network Monitor'. The main content area is titled 'Installation Mode' and contains the following text: 'Choose between the express or the custom installation mode.' Below this, there are two radio button options. The first option is 'Express (recommended)' and is selected. It includes a list of features: '- Use the default installation path and data path', '- Immediately run an auto-discovery of your network using standard protocols (ICMP, SNMP, and others)', '- Display all available devices after the application start', and '- Configure your monitoring setup further in the PRTG interface later'. The second option is 'Custom' and is not selected. It includes a list of features: '- Manually select the installation path and data path', '- Configure or skip the auto-discovery of your network', and '- Configure your monitoring setup further in the PRTG interface later'. At the bottom, there is a footer with 'www.paessler.com' and three buttons: 'Back', 'Next' (highlighted with a blue border), and 'Cancel'.

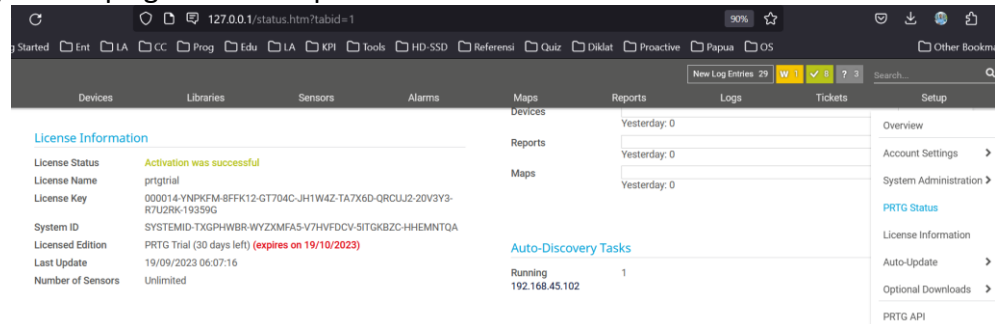
42) Silahkan lanjut instalasi sampai selesai



43) Jika instalasi sudah berhasil, maka akan muncul halaman sebagai berikut



44) Lisensi prtg bisa dilihat pada menu PRTG Status



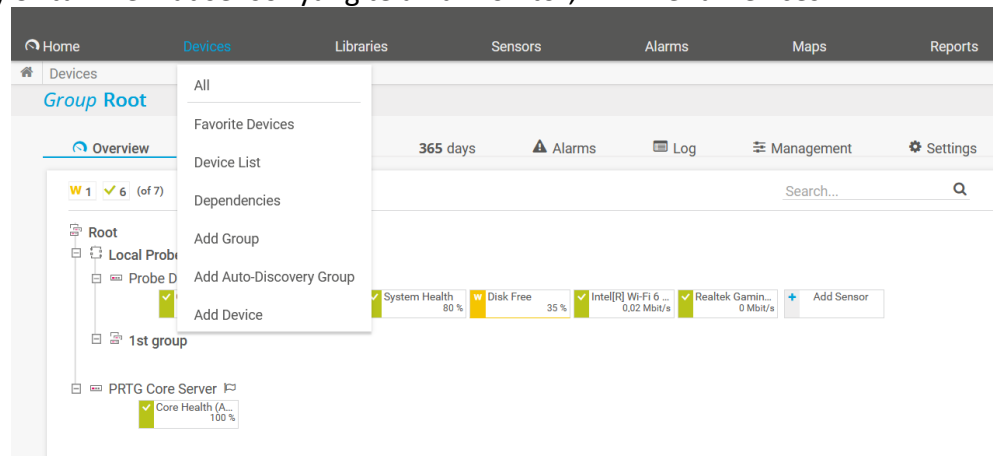
Jika trial sudah berakhir, maka prtg masih bisa digunakan tapi terbatas untuk 100 sensor.

Is PRTG Network Monitor free?

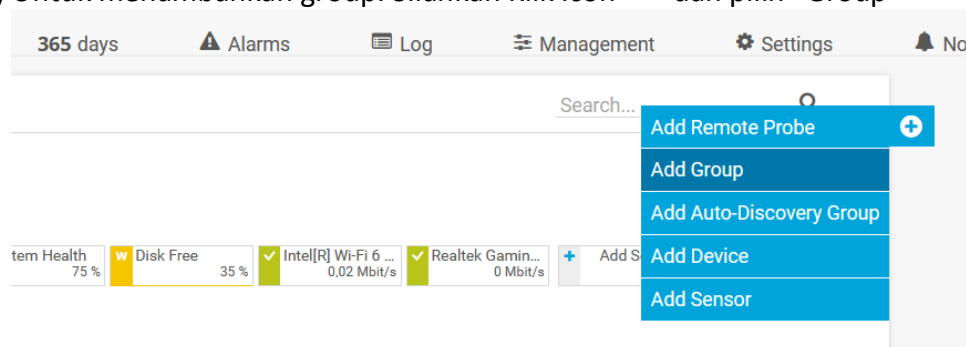
Paessler PRTG Network Monitor offers a free 30-day trial that includes all monitoring, alerting, and reporting features as well as historical data analysis. There's also a freeware edition of PRTG Network Monitor that is free of charge for up to 100 PRTG sensors (or monitored aspects). If your trial period expires or you need more than 100 sensors, you have to purchase one of the various perpetual PRTG licenses.

45) Server NMS sudah siap digunakan untuk memonitoring perangkat jaringan atau host/server yang ada dalam jaringan.

46) Untuk melihat sensor yang telah dimonitor, Klik Menu Devices



47) Untuk menambahkan group. Silahkan Klik Icon “+” dan pilih “Group”



Buat nama group, missal “Lab KPTK”

Add a Group to Local Probe

Group Name ⓘ

LAB KPTK

Tags ⓘ



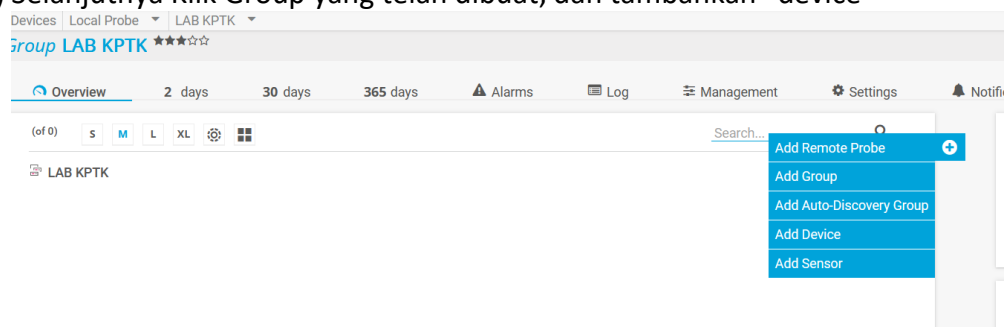
Credentials for Windows Systems



inherit from Local Probe

(Domain or Computer Name: <empty>, User Name

48) Selanjutnya Klik Group yang telah dibuat, dan tambahkan “device”



Misal, menambahkan router cisco dengan ip manajemen yang telah dibuat.

Add Device to Group LAB KPTK

Device Name and Address

Device Name ⓘ

Router Cisco - 100.100.1.1

IP Version ⓘ

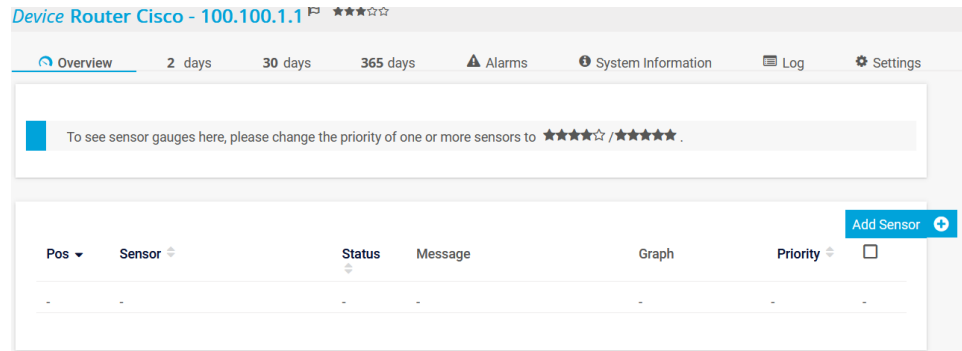
☒ IPv4

☐ IPv6

IPv4 Address/DNS Name ⓘ

100.100.1.1

Masuk kedalam Device “Router Cisco”, lalu tambahkan sensor. Misal sensor ping, sensor trafik, dan lain-lain.



Misalkan Sensor “ping”

Search  ping

Matching Sensor Types

Cloud Ping v2 ? Monitors the ping time to the target device from different locations on a number of continents <i>Requires access to the internet. The probe system must be able to communicate with the PRTG Cloud. The address in the parent device settings must be reachable over the internet. You cannot monitor localhost or target devices</i>	HTTP Transaction ? Monitors a web server by performing a transaction using a set of URLs <i>Monitors whether logins or shopping carts work properly.</i>	Ping ? Monitors connectivity using ping <i>Ping requests are used to check whether a device is reachable through the network.</i>
---	---	--

Add Sensor to Device Router Cisco - 100.100.1.1 [100.100.1.1]

< Cancel

Basic Sensor Settings

Sensor Name ⓘ Ping

Parent Tags ⓘ

Tags ⓘ pingsensor x +

Priority ⓘ ★★★★★

Menambahkan sensor snmp trafik

Search  snmp

Most Used Sensor Types

SNMP Traffic


Monitors bandwidth and traffic on servers, PCs, switches, etc. using SNMP

To query data from a probe device (localhost, 127.0.0.1, or ::1), add this device to PRTG with the IP address it has in your network and create the sensor on this device.




49) Contoh sensor yang sudah ditambahkan

Pos ▾	Sensor ▾	Status ▾	Message	Graph
1.	 Ping	Up	OK	
2.	 (003) TO ROUTER C...	Unus...	1 day interval average of 8...	
3.	 (009) 2012005930-...	Up	OK	
4.	 (010) 2017011167-...	Up	OK	

50) Untuk membuat dashboard monitoring, bisa menggunakan fitur “Map”

Devices
 Libraries
 Sensors
 Alarms
 Maps
 Reports
 Logs
 Tickets

ap

Map

Cancel

Map Name

Map Name  Map

Map Layout

Map Width  800

Map Height  600

Background Image (optional)  

51) Contoh Map yang telah dibuat

